Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
8	*(a)*	(i)	Use of $\lambda = \frac{hc}{\Delta E}$ including wrong transition **or** equivalent (1) 1.55 μ[m] (1)		2		2	2	
		(ii)	**Absorption:** photon disappears **or** is absorbed **or** has its energy transferred (1) Electron promoted <u>from L</u> (or ground state) [to U] (1) Don't accept if state going to P **Stimulated em:** [passing] photon demotes electron <u>from U</u> [to L] (1) [another] photon emitted; 2 photons must be implied (1)	4			4		
		(iii)	Stimulated emission because more electrons <u>in U</u> [than L] Accept population inversion		1		1		
		(iv)	So P stays relatively empty to receive [pumped] electrons **or** so as not to impede pumping **or** equivalent **or** so as to keep the population of U high (1) To maintain high population in U **or** to maintain population inversion **or** equivalent (1)	2			2		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)		**At least 1 from:** - no power loss in wires - high power lasers very inefficient - laser beam only travels in straight lines / wires are flexible - less use of raw materials (e.g. copper) for laser beams - no danger of electrocution with laser beams - light isn't versatile needs to be transferred into other more useful forms - reference to beam spreading and absorption **Maximum of 3 from the following:** - no visual pollution - cheaper because less raw materials used - replacement cost very large - danger to birds / aeroplanes, possibly to humans accept to anything getting in the way - if intercepted cuts power			4	4		
			Question 8 total	**6**	**3**	**4**	**13**	**2**	**0**